A Euclidean View of the Non-Homomorphism Factor: the Profile of a Graph

Mohammad Ali Rostami

Draft, October 26, 2026

Abstract

The non-homomorphism factor $|G, H|$ is the minimum number of edges that must be deleted from G so that the remaining graph admits a homomorphism to H. The symmetric quantity $\max(|G, H|, |H, G|)$ is not a metric, and its shortest-path closure is a metric that is not Euclidean. We study instead the *profile* of a graph G: the vector of normalised values $|T, G|/|E(T)|$ over a family of test graphs T. The profile depends only on the homomorphism class of G, reverses the homomorphism order, and yields an exact Euclidean metric. We give closed forms for its clique and odd-cycle coordinates, which lead to a strengthening of the clique lower bound for $|G, H|$. We show that no family of bounded-size tests can detect the chromatic number, and relate the profile to the separation parameter of Färnqvist, Jonsson and Thapper and to the concentration parameter. Computations on all 1276 cores with at most 9 vertices, on the generalized Petersen graphs $GP(n, k)$ with $n \leq 12$, and on the circular cliques $K_{p/q}$ with $p \leq 13$, illustrate the geometry of the resulting space.

1 Introduction

For graphs G and H, the *non-homomorphism factor* $|G, H|$ is the minimum number of edges whose removal from G leaves a graph that maps homomorphically to H [7]. Deciding whether $|G, H| \leq k$ is the problem Edge Deletion-H-Colouring [5]. It is NP-hard whenever the core of H has more than one edge, by the Hell–Nešetřil dichotomy [6].

A natural attempt to measure how far apart two graphs are is $\operatorname{dist}(G, H) = \max(|G, H|, |H, G|)$. It is symmetric and, on cores, vanishes only on isomorphic pairs. However, it is not a metric:

$$\operatorname{dist}(K_2, K_5) = 4 \;>\; 3 = \operatorname{dist}(K_2, K_3) + \operatorname{dist}(K_3, K_5).$$

The failure is unbounded. The chain $K_n, K_{n-1}, \ldots, K_2$ has total length $n-2$, while $\operatorname{dist}(K_n, K_2) \approx n^2/4$. The largest metric below dist is its shortest-path closure d^*, for which $d^*(K_n, K_m) = |n - m|$. But d^* cannot be embedded isometrically in any Euclidean space. Indeed, $d^*(K_3, K_4) = d^*(K_4, K_5) = d^*(K_4, I) = 1$, where I is the icosahedron, while the three distances among K_3, K_5 and I all equal 2. In a Euclidean space, K_4 would have to be the midpoint of all three pairs, which is impossible.

This paper takes a different route. Instead of comparing two graphs directly, we describe each graph by how a fixed family of test graphs maps *into* it. This gives the *profile* of a graph, and ordinary Euclidean distance between profiles.

Throughout, $\operatorname{mc}_G(T) = |E(T)| - |T, G|$ denotes the largest number of edges of T preserved by a vertex map $V(T) \to V(G)$. This is the optimum of the Max G-Col problem [4, 2], also called maximum graph homomorphism [8]. We write $t(n, r)$ for the number of edges of the Turán graph $T(n, r)$.

Several of the statements below are known in the language of the *separation* parameter of Engström, Färnqvist, Jonsson and Thapper [2]; we indicate this where it applies. What we add is the Euclidean embedding, the improved lower bound (Remark 3.2), the blindness result (Proposition 3.4), the link to the concentration parameter (Proposition 5.2), and the computations of Section 6.

2 The profile and the profile metric

Definition 2.1 (Profile). *Let $\mathfrak{F}$ be a countable family of graphs, each with at least one edge (the* test graphs*). For a graph G with at least one edge, the* profile *of G is*

$$\varphi(G) = \big(\varphi_T(G)\big)_{T\in\mathfrak{F}}, \qquad \varphi_T(G) = \frac{|T,G|}{|E(T)|} \in [0,1].$$

Given weights $w\colon \mathfrak{F} \to (0,\infty)$ with $\sum_T w_T < \infty$, the profile distances *are*

$$E_w(G,H) = \Big(\sum_{T\in\mathfrak{F}} w_T\big(\varphi_T(G) - \varphi_T(H)\big)^2\Big)^{1/2}, \qquad E_\infty(G,H) = \sup_{T\in\mathfrak{F}} \big|\varphi_T(G) - \varphi_T(H)\big|.$$

E_w is the distance of the Hilbert space $\ell^2(\mathfrak{F}, w)$ restricted to the set of profiles, so it is Euclidean by construction. The profile is the target-side, max-plus analogue of Lovász's vector of homomorphism numbers $(\hom(T,G))_T$ [9, 1]. Its zero set $\{T : \varphi_T(G) = 0\} = \{T : T \to G\}$ is the qualitative version of that vector. Unlike the source-side quantity $|G,H|$, the profile depends only on the homomorphism class of G. For example, $|2K_4, K_3| = 2 \neq 1 = |K_4, K_3|$, although $2K_4$ and K_4 are homomorphically equivalent.

Proposition 2.2 (Monotonicity and order embedding). *(a) If $G \to H$, then $\varphi_T(G) \geq \varphi_T(H)$ for every T. In particular, $\varphi(G) = \varphi(H)$ whenever G and H are homomorphically equivalent.*

(b) If $\mathfrak{F}$ contains a graph homomorphically equivalent to G, then $\varphi(G) \geq \varphi(H)$ implies $G \to H$.

Consequently, if $\mathfrak{F}$ contains all connected cores, then $G \mapsto \varphi(G)$ is an order-reversing embedding of the homomorphism order of connected graphs into $([0,1]^{\mathfrak{F}}, \leq)$. In that case E_w and E_∞ are metrics on homomorphism classes, and for finite $\mathfrak{F}$ the sum $\|\varphi(G)\|_1$ is a strictly decreasing height function on the cores in $\mathfrak{F}$.

Proof. (a) If $T - S \to G \to H$, then $T - S \to H$. This is the profile form of [2, Lemma 2.6]. (b) Let $T \in \mathfrak{F}$ with $T \leftrightarrow G$. Then $0 = \varphi_T(G) \geq \varphi_T(H)$, so $G \to T \to H$. □

Part (b) needs a graph equivalent to G itself; a graph equivalent to H is not enough. For example, with $\mathfrak{F} = \{K_2\}$ we have $\varphi_{K_2}(K_3) = 0 = \varphi_{K_2}(K_2)$, so $\varphi(K_3) \geq \varphi(K_2)$, although $K_3 \not\to K_2$.

The embedding is not a lattice map. The join and meet of the homomorphism order are $G \sqcup H$ and $G \times H$. They satisfy $\varphi(G \sqcup H) \leq \min(\varphi(G), \varphi(H))$ and $\varphi(G \times H) \geq \max(\varphi(G), \varphi(H))$, and both inequalities can be strict. For example, let T be K_3 and the Grötzsch graph Gr joined by a bridge. Then $|T, K_3| = |T, \mathrm{Gr}| = 1$, but $|T, K_3 \times \mathrm{Gr}| = 2$.

3 Closed forms and what the profile can read

Proposition 3.1 (Clique and odd-cycle coordinates). *Let G have $\omega(G) = r \geq 2$ and odd girth g, with $g = \infty$ if G is bipartite. For every $n \geq 2$ and $k \geq 1$,*

$$|K_n, G| = \binom{n}{2} - t\big(n, \min(n,r)\big), \qquad \varphi_{C_{2k+1}}(G) = \begin{cases} 0, & 2k+1 \geq g, \\ \dfrac{1}{2k+1}, & 2k+1 < g. \end{cases}$$

Every bipartite test has $\varphi_T(G) = 0$. Hence $\omega(G)$ and the odd girth of G can be read off $\varphi(G)$: $\omega(G) \geq n \iff \varphi_{K_n}(G) = 0$, and $g \leq 2k+1 \iff \varphi_{C_{2k+1}}(G) = 0$.

Proof. The preserved edges of K_n form a graph that maps to G. Cliques map injectively, so this graph is K_{r+1}-free, and by Turán's theorem it has at most $t(n,r)$ edges. Mapping K_n in a balanced way onto an r-clique of G attains this bound. For odd cycles, $C_{2k+1} \to G$ if and only if G has an odd cycle of length at most $2k+1$. Otherwise, deleting one edge leaves a path, which maps to an edge. In the language of separation these are [2, Props. 2.4, 2.5]. □

Remark 3.2 (An ω-version of the clique lower bound). *The same argument gives, for all graphs G and H,*

$$|G,H| \;\geq\; |E(G)| - t\big(|V(G)|, \omega(H)\big),$$

because every subgraph of G that maps to H is $K_{\omega(H)+1}$-free. Since $\omega(H) \leq \chi(H)$, this is never weaker than the bound $|G,H| \geq |E(G)| - t(|V(G)|, \chi(H))$, and it is attained by $G = K_n$ for every *H. For example, $|K_5, C_5| = 4$: the largest triangle-free subgraph of K_5 is $K_{2,3}$, which maps to C_5. The χ-bound gives only $10 - t(5,3) = 2$.*

Proposition 3.3 (Range of the profile)**.** *For every G with an edge and every test T, $\varphi_T(G) \le \varphi_T(K_{\omega(G)}) \le 1/\omega(G)$. Moreover, $\varphi_{K_n}(G) \to 1/\omega(G)$ as $n \to \infty$, so $\sup_T \varphi_T(G) = 1/\omega(G)$. In particular, $E_\infty(G,H) \ge \bigl|1/\omega(G) - 1/\omega(H)\bigr|$ when $\mathfrak{F}$ contains all complete graphs.*

Proof. The first inequality is Proposition 2.2(a) applied to $K_{\omega(G)} \to G$. For the second, colour $V(T)$ uniformly at random with $\omega(G)$ colours. Each edge is properly coloured with probability $1 - 1/\omega(G)$, so some colouring preserves at least that fraction of $E(T)$. The limit follows from Proposition 3.1. □

Proposition 3.4 (Bounded tests cannot see χ)**.** *Suppose every $T \in \mathfrak{F}$ has at most m vertices, and G has an edge and odd girth larger than m. Then $\varphi(G) = \varphi(K_2)$ on $\mathfrak{F}$.*

Proof. Let $f\colon V(T) \to V(G)$. The image of the preserved edges is a subgraph of G on at most m vertices. It has no odd cycle, so it maps to an edge of G. Hence $\mathrm{mc}_G(T) = \mathrm{mc}_{K_2}(T)$. □

By Erdős' theorem on graphs of large girth and large chromatic number [3], no finite family of tests detects χ, the fractional chromatic number, or even bipartiteness. With summable weights, $E_w(G, K_2) \to 0$ along any sequence of graphs whose odd girth tends to infinity, whatever their chromatic numbers. Concretely, the generalized Mycielskian $\mu_5(C_{11})$ (67 vertices, $\chi = 4$) has the same profile as K_2 on the twelve tests $K_2, \dots, K_6$, C_5, C_7, C_9, Petersen, Grötzsch, W_5 and the icosahedron. Section 6 shows the same effect on generalized Petersen graphs.

Circular cliques show the mechanism precisely. Recall that $K_{p/q}$ ($\gcd(p,q) = 1$, $p \ge 2q$) has vertex set $\mathbb{Z}_p$, with $i \sim j$ if and only if $q \le |i-j| \le p-q$; then $K_{n/1} = K_n$ and $K_{(2k+1)/k} = C_{2k+1}$. A graph X with an edge maps to $K_{p/q}$ if and only if its circular chromatic number satisfies $\chi_c(X) \le p/q$, and $\chi_c(X) = a/b$ for some $a \le |V(X)|$ [10, 11]. For a rational $r \ge 2$, let $\lfloor r \rfloor_m$ be the largest fraction $a/b \le r$ with $a \le m$.

Proposition 3.5 (Farey staircase)**.** *If every $T \in \mathfrak{F}$ has at most m vertices, then $\varphi(K_{p/q}) = \varphi\bigl(K_{\lfloor p/q \rfloor_m}\bigr)$ on $\mathfrak{F}$.*

Proof. Let $T \in \mathfrak{F}$ and $S \subseteq E(T)$. The graph $T - S$ has at most m vertices, so $\chi_c(T-S)$ is a fraction with numerator at most m. Hence $\chi_c(T-S) \le p/q$ if and only if $\chi_c(T-S) \le \lfloor p/q \rfloor_m$, that is, $T - S \to K_{p/q}$ if and only if $T - S \to K_{\lfloor p/q \rfloor_m}$. So $|T, K_{p/q}| = |T, K_{\lfloor p/q \rfloor_m}|$. □

Proposition 3.4 restricted to odd cycles is the case $\lfloor (2k+1)/k \rfloor_m = 2$ for $2k+1 > m$. Along the chain of circular cliques, which is a chain in the homomorphism order, the profile is therefore a staircase: it only moves at fractions with small numerators.

4 The source-side profile

The profile records what maps *into* G. The chromatic number is a statement about where G maps *to*, so it calls for the dual object. The naive candidate $|G,T|/|E(G)|$ is not constant on homomorphism classes, so we use weights.

Definition 4.1. *For a graph G with an edge and a test T, let*

$$\psi_T(G) = \max_w \frac{|G,T|_w}{\|w\|_1},$$

where the maximum is over non-negative, non-zero edge weights w on G, and $|G,T|_w$ is the minimum total weight of edges whose deletion leaves a graph mapping to T.

Proposition 4.2. *$\psi_T(G) = 1 - s(T,G)$, where s is the separation of Färnqvist, Jonsson and Thapper [4, 2]. Moreover:*

(a) ψ_T is constant on homomorphism classes and order-preserving: $G \to H$ implies $\psi_T(G) \le \psi_T(H)$.

(b) $\psi_T(G) = 0$ if and only if $G \to T$. In particular $\psi_{K_k}(G) > 0$ if and only if $\chi(G) > k$.

(c) $\psi(G \sqcup H) = \max(\psi(G), \psi(H))$ and $\psi(G \times H) \le \min(\psi(G), \psi(H))$.

(d) $|G,T|/|E(G)| \le \psi_T(G)$, with equality if G is edge-transitive.

Sketch. The identity with $1-s(T,G)$ is [2, Thm. 2.1, Cor. 2.11], which also shows the maximum is attained by automorphism-invariant weights on G itself. By [2, Prop. 5.4] it equals $1-1/\chi_T(G)$, where χ_T is the fractional covering number by T-cuts; for $T=K_2$ this is Šámal's cubical chromatic number [12]. (a)–(c) follow from the definition, and (d) from averaging over automorphisms. □

Thus φ reads the clique number and ψ reads the chromatic number. The gap $\chi-\omega$ appears as the distance between the two readings; see Figure 2(b). Hedetniemi's conjecture asserted that $\chi(G\times H)>k$ whenever $\chi(G)>k$ and $\chi(H)>k$. In profile terms, it asserted that $\psi_{K_k}(G)>0$ and $\psi_{K_k}(H)>0$ imply $\psi_{K_k}(G\times H)>0$. Shitov's counterexamples [13] show that this fails, so by Proposition 4.2(c) the inequality $\psi(G\times H)\le\min(\psi(G),\psi(H))$ can be strict even on the support.

5 Three distances on one embedding

Proposition 5.1 (Relation to separation)**.** *For edge-weighted graphs (T,w), write $p_G(T,w)=\mathrm{mc}_G(T,w)/\|w\|_1$; for unit weights, $p_G(T)=1-\varphi_T(G)$. The separation of [4, 2] is $s(G,H)=\inf_{(T,w)}p_G(T,w)/p_H(T,w)$.*

(a) If T is edge-transitive, then $\varphi_T(G)=1-s(G,T)$ [2, Thm. 2.1].

(b) The additive form of the FJT metric is the Hilbert projective metric on profiles:

$$-\log s(G,H)-\log s(H,G)=\log\sup_{(T,w)}\frac{p_H(T,w)}{p_G(T,w)}+\log\sup_{(T,w)}\frac{p_G(T,w)}{p_H(T,w)},$$

and the FJT distance is $d_{\mathrm{FJT}}=1-s(G,H)s(H,G)=1-e^{-d_{\mathrm{Hilb}}}$. For a finite unweighted family $\mathfrak{F}$, the same expression with the suprema over $T\in\mathfrak{F}$ is a pseudometric. It is a metric on the homomorphism classes of graphs whose cores lie in $\mathfrak{F}$.

Proof. (a) is the edge-transitive case of [2, Thm. 2.1]. (b) is the definition of s, rewritten. Positivity on a finite family follows as in Proposition 2.2(b). □

So the sup-distance E_∞, the Euclidean distance E_w and the FJT metric are three distances on one embedding $G\mapsto\varphi(G)$. Only the FJT metric carries an approximation-theoretic meaning [2, Lemma 1.6]; only E_w is Euclidean.

Proposition 5.2 (Moving the target; link to concentration)**.** *Let $S\subseteq E(T)$ with $|S|=|T,G|$, and let $f\colon T-S\to G$ be a homomorphism. Let $\mu(f)$ be the largest number of edges of $T-S$ that f maps onto a single edge of G. For every graph H,*

$$|T,H|\ \le\ |T,G|+\mu(f)\,|G,H|.$$

In particular, if $T\to G$, then $|T,H|\le\gamma(T,G)\,|G,H|$, where γ is the concentration parameter. Also $|T,G-e|\le|T,G|+\mu(f,e)$ for every $e\in E(G)$, where $\mu(f,e)$ is the load of e.

Proof. Let $S'\subseteq E(G)$ be optimal with $g\colon G-S'\to H$. Delete from $T-S$ the at most $\mu(f)|S'|$ edges that f maps into S'. The map $g\circ f$ is then a homomorphism from what remains to H. □

The same inequality, with T replaced by G, is what breaks the triangle inequality for dist: the composition cost is multiplied by the load. The concentration parameter is therefore the local Lipschitz constant of the profile under changes of the target. The jump can be large: deleting a triangle edge from K_3 with a long pendant path changes φ_{K_3} from 0 to 1/3, however many edges the graph has.

Remark 5.3 (Computation)**.** *We have*

$$\mathrm{mc}_G(T)=\lim_{\beta\to\infty}\beta^{-1}\log\hom(T,W_\beta),$$

where W_β is the looped complete graph on $V(G)$ with weight e^β on $E(G)$ and 1 elsewhere. Consequently, the tree-decomposition dynamic program for $\hom(T,\cdot)$, run over the $(\max,+)$ semiring, computes $\varphi_T(G)$ in time $O\big(|V(T)|\,|V(G)|^{\mathrm{tw}(T)+1}\big)$.

6 Computations

Data. All values below are exact. We computed $|T, G|$ for all pairs among the 1276 cores of connected graphs on at most 9 vertices (plus a few named graphs), with a branch-and-bound solver. The default test family is the set of 85 cores consisting of all cores on at most 8 vertices together with C_9, C_{11}, the Petersen graph, the Grötzsch graph and the icosahedron. Each test T receives the weight $w_T = 1/\sqrt{N_{\chi(T)}}$, where N_c is the number of tests with chromatic number c, so that each chromatic class of tests carries the same total squared weight. With these weights E_w correlates 0.92 with dist and 0.83 with d^* (Pearson), better than with uniform weights (0.78 and 0.59). Removing any single test changes the ranking of distances very little (Spearman $\rho \geq 0.9995$).

Geometry. Figure 1 shows the first two principal components. The first axis carries 92% of the variance and is, up to sign, the height $\|\varphi\|_1$ of Proposition 2.2. Along the 615,167 comparable pairs $G \to H$ among the 1276 cores it never decreases. The complete graphs form a chain $K_8, K_7, \ldots, K_3$ that then jumps to K_2 at the triangle-free end, where the odd cycles accumulate: $C_{2k+1} \to K_2$ in the profile metric, as Proposition 3.1 predicts. Nearest neighbours share the clique number in 96% of cases but the chromatic number in only 81%; this agrees with Proposition 3.1 and Proposition 3.4.

Generalized Petersen graphs. We computed the full 85-test profile of all 30 graphs $GP(n, k)$ with $3 \leq n \leq 12$ and $1 \leq k < n/2$ (Table 1). Every one of them lands *exactly* on the profile of one of the seven cores K_2, K_3, C_5, C_7, C_9, C_{11} or Petersen. For 19 of them, including $GP(5, 2)$, which is the Petersen graph itself, this is because they are homomorphically equivalent to that core. The other 11 are 3-chromatic graphs of odd girth 5 or 7 that do *not* map to C_5 or C_7, as an exhaustive search confirms. Yet no test with at most 12 vertices distinguishes them from the cycle. The dodecahedron $GP(10, 2)$ is such a graph.

Profile equals that of	Homomorphically equivalent	Same profile, *not* equivalent
K_2	$GP(4,1)$, $GP(6,1)$, $GP(8,1)$, $GP(8,3)$, $GP(10,1)$, $GP(10,3)$, $GP(12,1)$, $GP(12,3)$, $GP(12,5)$	—
K_3	$GP(3,1)$, $GP(6,2)$, $GP(9,3)$, $GP(12,4)$	—
C_5	$GP(5,1)$, $GP(10,4)$	$GP(7,2)$, $GP(7,3)$, $GP(8,2)$, $GP(9,2)$, $GP(9,4)$, $GP(10,2)$, $GP(11,2)$, $GP(11,5)$, $GP(12,2)$
C_7	$GP(7,1)$	$GP(11,3)$, $GP(11,4)$
C_9	$GP(9,1)$	—
C_{11}	$GP(11,1)$	—
Petersen	$GP(5,2)$	—

Table 1: Where the generalized Petersen graphs $GP(n, k)$, $3 \leq n \leq 12$, land in the profile space of the 85 test cores. The left column lists graphs that are homomorphically equivalent to the core (bipartite, or containing a triangle, or mapping onto their shortest odd cycle). The right column lists graphs with exactly the same profile but no homomorphism to the cycle.

Circular cliques. We also computed the profiles of all 24 circular cliques $K_{p/q}$ with $p \leq 13$ and $2 \leq p/q \leq 8$. Thirteen of them are themselves test graphs ($K_2, \ldots, K_8$, C_5, C_7, C_9, C_{11}, $K_{7/2}$, $K_{8/3}$). Each of the other eleven lands exactly on the point of the largest circular clique below it whose numerator is at most 8. For example, $K_{13/6}$ lands on K_2, $K_{12/5}$ on $C_7 = K_{7/3}$, $K_{13/5}$ on C_5, $K_{11/4}$ on $K_{8/3}$, $K_{10/3}$ and $K_{13/4}$ on K_3, $K_{11/3}$ on $K_{7/2}$, and $K_{9/2}$ on K_4. This is the staircase of Proposition 3.5, with an even coarser resolution than the largest test (12 vertices) guarantees. In particular, $K_{10/3}$ has chromatic number 4 but the same profile as K_3.

Separating families. On the 1276 cores, every test family that separates all homomorphism classes has at least 253 members; a greedy search finds one with 311. The reason is structural. Suppose $H \to G$, and $G - e \to H$ for every edge e of G (a *critical cover*). Then a test T separates G from H only if every optimal map $T \to G$ uses every edge of G, which in practice forces $T \leftrightarrow G$.

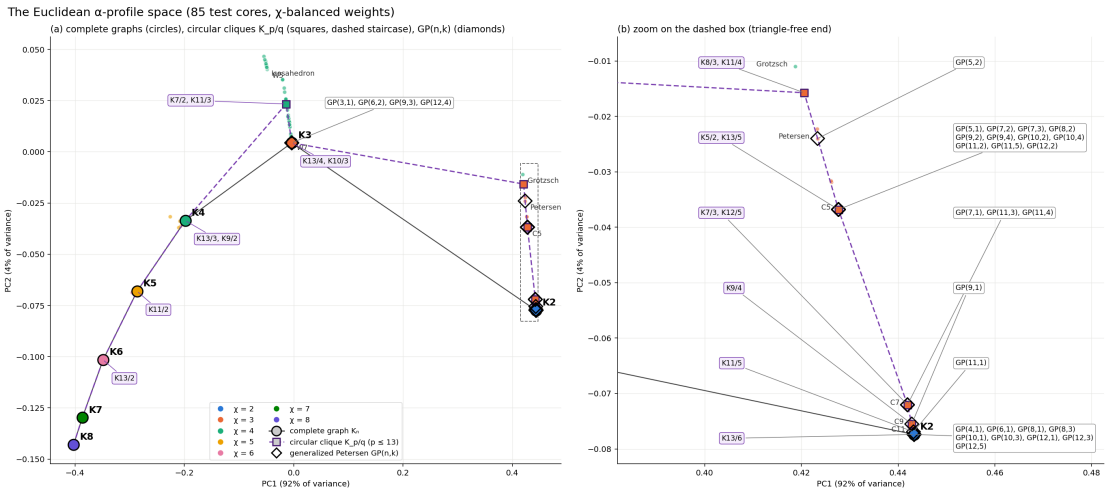


Figure 1: The Euclidean profile space of the 85 test cores (χ-balanced weights), first two principal components. Large circles: complete graphs, joined in order. Squares joined by a dashed line: circular cliques $K_{p/q}$, $p \le 13$, in order of p/q. Diamonds: generalized Petersen graphs $GP(n,k)$, $n \le 12$. Panel (b) zooms on the triangle-free end, where the odd cycles accumulate at K_2.

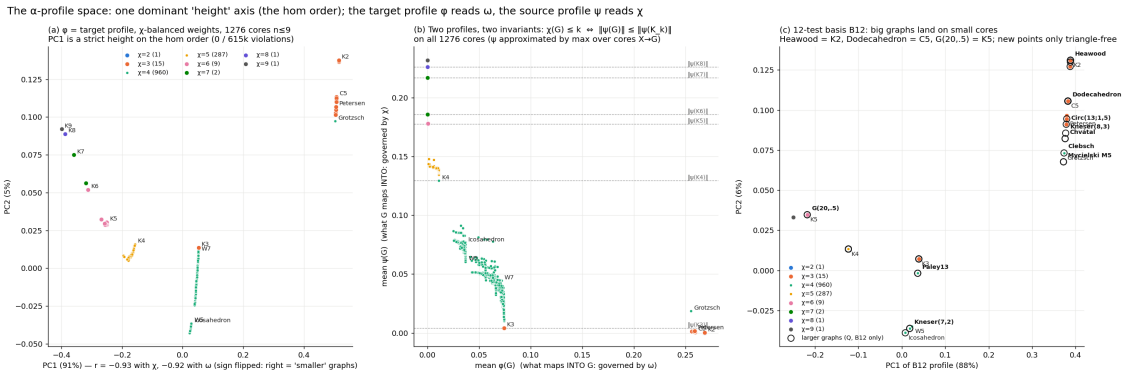


Figure 2: (a) The 1276 cores on at most 9 vertices; the first principal component is a strict height on the homomorphism order. (b) Mean of φ (what maps into G, governed by ω) against mean of ψ (where G maps, governed by χ); here ψ is approximated by the maximum of $|X,T|/|E(X)|$ over cores $X \to G$ in the data, which includes $X = G$. (c) Larger graphs on a 12-test basis land on small cores; only triangle-free graphs of larger chromatic number give new points.

Of the 253 forced pairs, 251 are critical covers. Adding the source-side coordinates ψ_H removes this obstruction.

7 Open questions

Question 1 (Size of a separating family)**.** *Let $\sigma(n)$ be the least size of a family $\mathfrak{F}$ such that φ restricted to $\mathfrak{F}$ separates all cores on at most n vertices. We know $253 \le \sigma(9) \le 311$. Is $\sigma(n)$ comparable to the number of cores on n vertices? Which cores are forced?*

Question 2 (Closure of the profile space)**.** *The image of φ is not closed: $\varphi(K_n) \to 0$, and the Mycielski chain converges to a point that is not the profile of a finite graph. Is the closure of $\{\varphi(G)\}$ in the product topology exactly the set of profiles of countable graphs? Is the space of homomorphism classes totally bounded under E_∞?*

Question 3 (Unweighted source-side profile)**.** *Is $\sup_{X \to G} |X,T|/|E(X)| = \psi_T(G)$ for all G and T, with the supremum over simple graphs X? Equality holds when G is edge-transitive.*

Question 4 (Is d^* quasi-Euclidean?)**.** *On cores with at most 9 vertices, d^* differs from $|\Delta\chi|$ by at most 3, yet $d^* \ge |\Delta\chi_f|$ suggests unbounded layers in general. Does d^* admit a bi-Lipschitz embedding into ℓ^2, or into the profile space?*

Acknowledgement of method

The computations and parts of the arguments were developed with the help of AI agents. Propositions marked with citations are known; the remaining statements have been checked on the data described above and should be read as drafts until independently verified.

References

[1] H. Dell, M. Grohe and G. Rattan, "Lovász meets Weisfeiler and Leman," in *ICALP 2018*, LIPIcs vol. 107, pp. 40:1–40:14, 2018.

[2] R. Engström, T. Färnqvist, P. Jonsson and J. Thapper, "An approximability-related parameter on graphs – properties and applications," *Discrete Mathematics & Theoretical Computer Science*, vol. 17, no. 1, pp. 33–66, 2015.

[3] P. Erdős, "Graph theory and probability," *Canadian Journal of Mathematics*, vol. 11, pp. 34–38, 1959.

[4] T. Färnqvist, P. Jonsson and J. Thapper, "Approximability distance in the space of H-colourability problems," in *CSR 2009*, LNCS vol. 5675, pp. 92–104, 2009. arXiv:0802.0423.

[5] F. Foucaud, H. Hocquard, D. Lajou, V. Mitsou and T. Pierron, "Parameterized complexity of edge-coloured and signed graph homomorphism problems," in *IPEC 2019*, LIPIcs vol. 148, pp. 15:1–15:16, 2019.

[6] P. Hell and J. Nešetřil, *Graphs and Homomorphisms*, Oxford University Press, 2004.

[7] K. Khoshkha, *Nonhomomorphism Factors*, M.Sc. thesis, Sharif University of Technology, 2005.

[8] M. Langberg, Y. Rabani and C. Swamy, "Approximation algorithms for graph homomorphism problems," in *APPROX 2006*, LNCS vol. 4110, pp. 176–187, 2006.

[9] L. Lovász, "Operations with structures," *Acta Mathematica Academiae Scientiarum Hungaricae*, vol. 18, pp. 321–328, 1967.

[10] A. Vince, "Star chromatic number," *Journal of Graph Theory*, vol. 12, no. 4, pp. 551–559, 1988.

[11] X. Zhu, "Circular chromatic number: a survey," *Discrete Mathematics*, vol. 229, pp. 371–410, 2001.

[12] R. Šámal, "Cubical coloring – fractional covering by cuts and semidefinite programming," *Discrete Mathematics & Theoretical Computer Science*, vol. 17, no. 2, 2015.

[13] Y. Shitov, "Counterexamples to Hedetniemi's conjecture," *Annals of Mathematics*, vol. 190, no. 2, pp. 663–667, 2019.